INTELLIGENT SYSTEMS • MOBILE • AI

Intelligent Cognitive Alarm Platform

Full Stack Development Progress

Presented By: **Jothiesh N** | Team 3

Project Overview

Intelligent Cognitive Alarm Platform (ICAP)

Objective

ICAP is an AI-powered mobile application designed to improve users' wake-up habits. Instead of allowing users to dismiss alarms directly, the app requires them to complete cognitive challenges before the alarm stops. The system also includes role-based access for Users, Wellness Coaches, and Administrators to monitor and improve user productivity and sleep habits.

Cognitive Alarms

Challenges gate alarm dismissal

Role-Based Access

User, Coach & Admin roles

Wellness Tracking

Productivity & sleep insights

My Contribution

My Role: Full Stack Developer

- React Native (Expo) mobile app development
- Built the FastAPI backend
- Implemented JWT-based authentication
- Role-based login: User, Coach & Administrator
- Integrated frontend with backend REST APIs
- Developed User Profile module
- Developed Alarm Scheduling module
- Developed Coach Dashboard
- Developed Administrator Dashboard
- Managed project architecture & navigation

Technology Stack

Frontend

- React Native (Expo)
- TypeScript
- Redux Toolkit
- React Navigation
- Axios

Backend

- Python
- FastAPI
- SQLAlchemy
- JWT Authentication
- REST APIs

Database

- PostgreSQL
- MongoDB

Developing by Database Team

AI / ML

- Cognitive Challenge Engine

Developing by AI/ML Team

Backend Development

FastAPI Backend Implementation — Completed Modules

- FastAPI Project Setup
- JWT Authentication
- User Registration
- User Login
- Refresh Token
- Forgot Password
- Reset Password
- Email Verification
- Current User API
- Role-Based Authentication
- REST API Development
- Secure Password Hashing
- API Testing using Swagger

Frontend Development

React Native Application

Developed Screens

- Splash Screen
- Login Screen
- Registration Screen
- Forgot Password Screen
- User Profile
- Alarm Scheduling
- Coach Dashboard
- Administrator Dashboard

Features

- Redux State Management
- Navigation
- API Integration
- Responsive UI
- Reusable Components

System Architecture

React Native (Expo)
Mobile Client

Axios API
HTTP Client Layer

FastAPI Backend
REST API Layer

PostgreSQL Database
Database Team

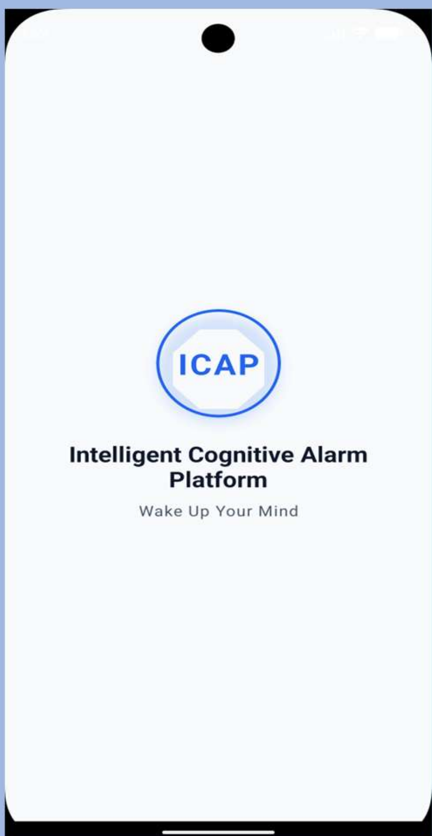
AI Cognitive Challenge
AI/ML Team

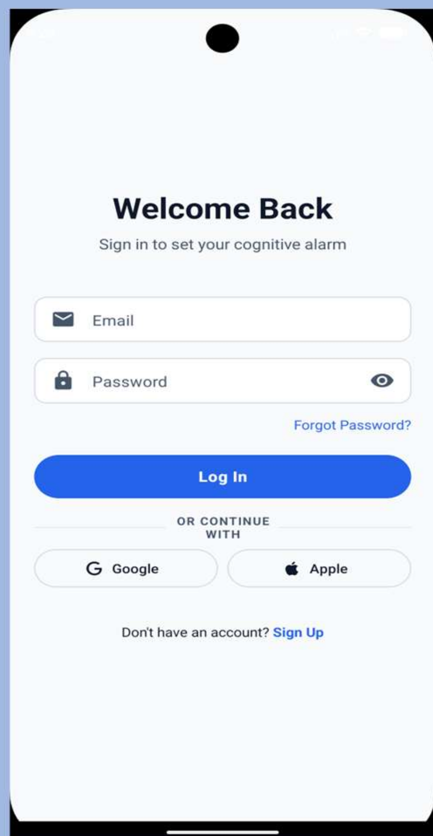
Integration

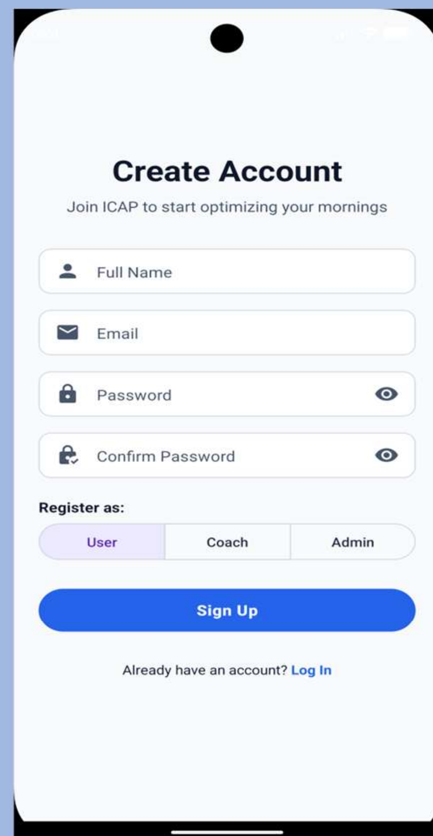
- Frontend communicates with FastAPI using REST APIs.
- FastAPI interacts with PostgreSQL for data storage.
- AI challenge module will be integrated through backend APIs.

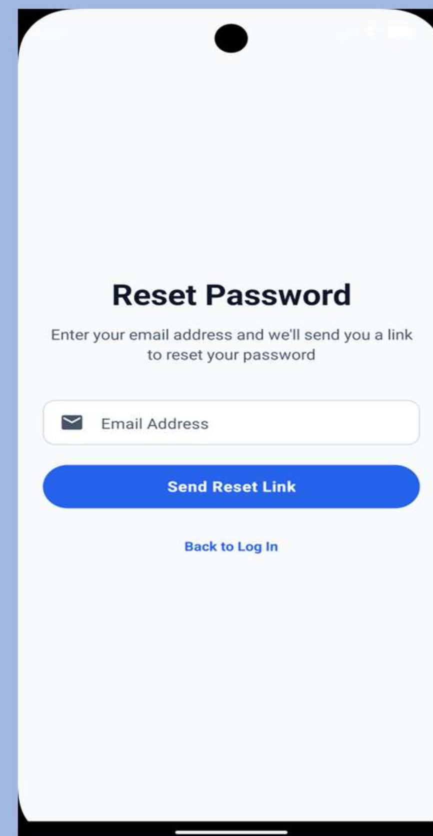
SnapShots:

Opening Screens

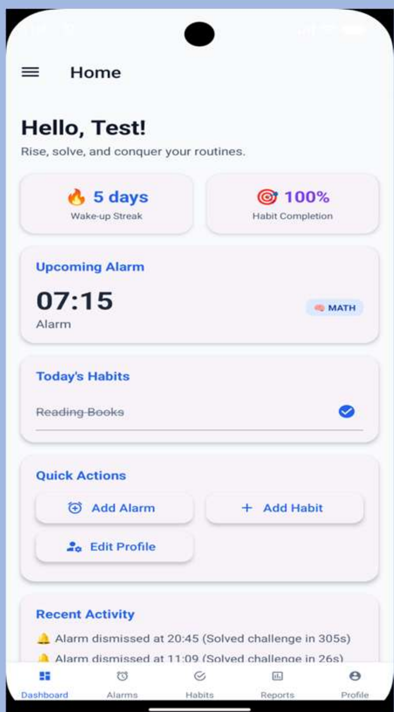


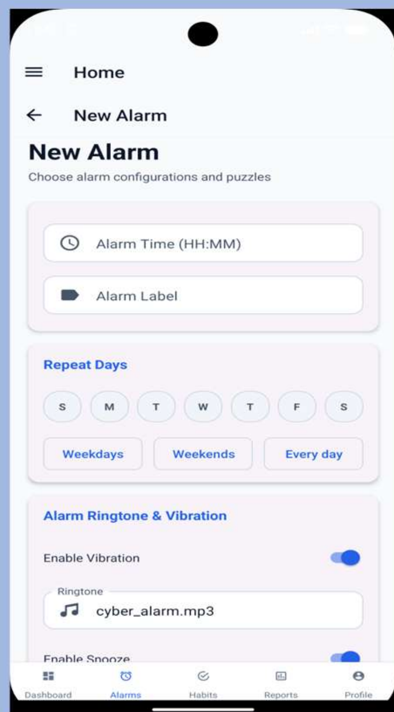


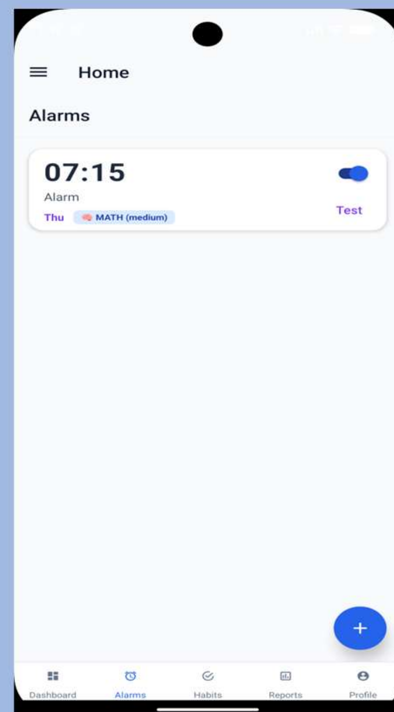


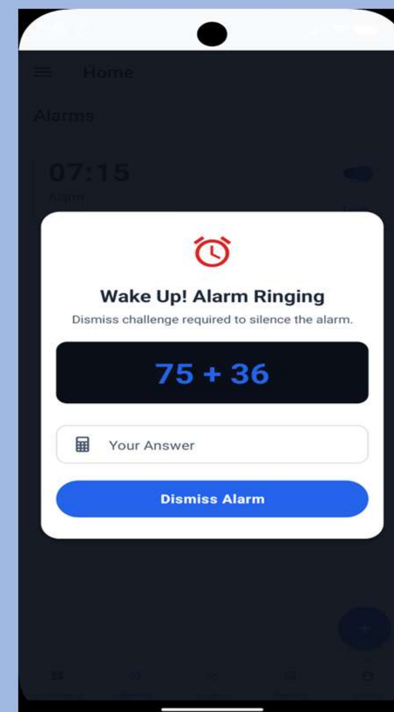


User Screens

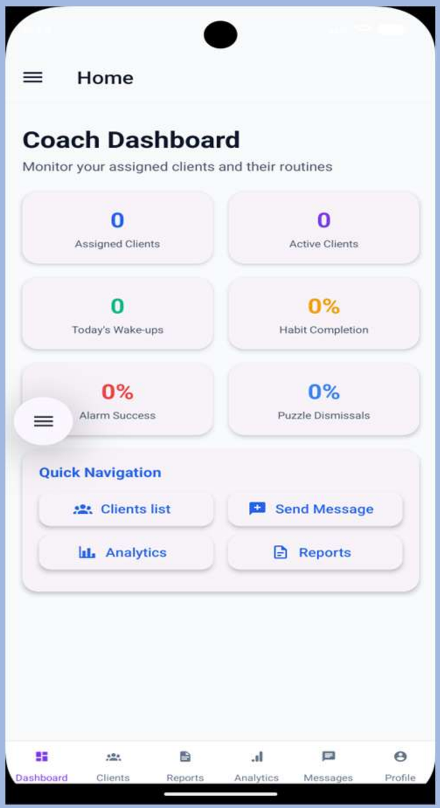


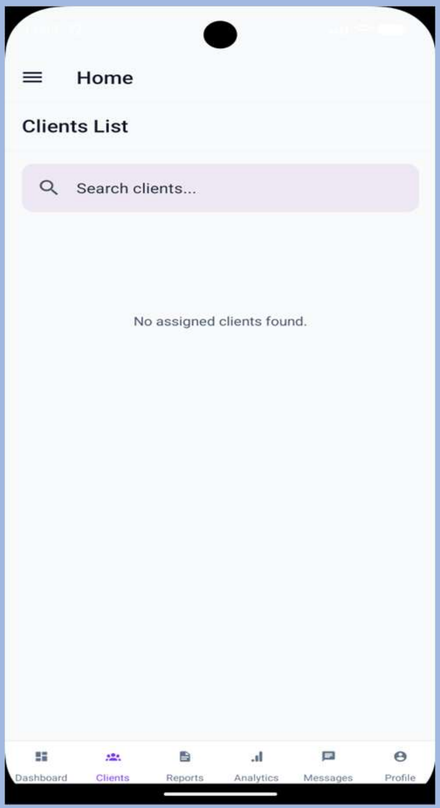


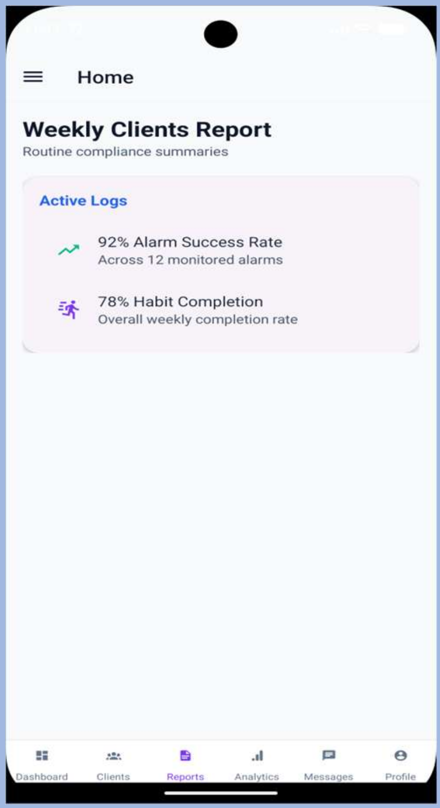


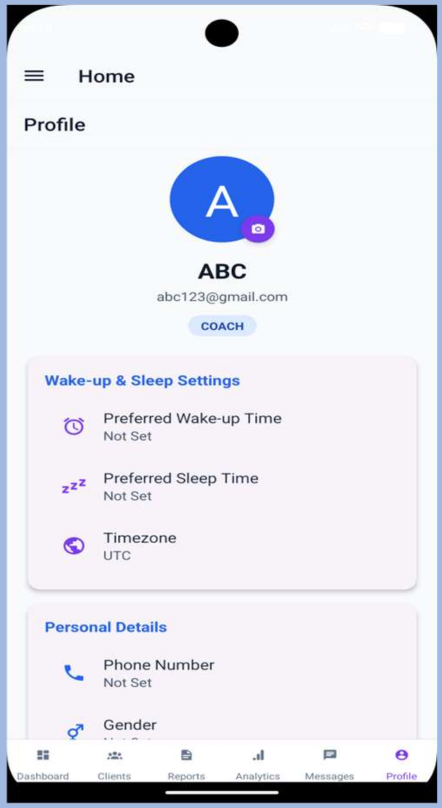


Coach Screens

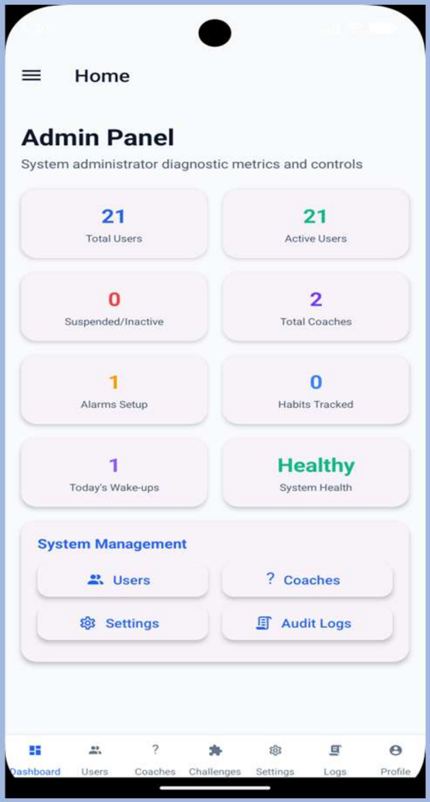


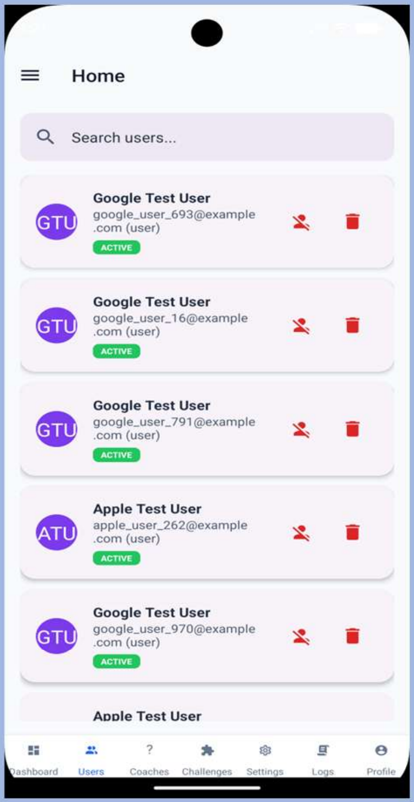


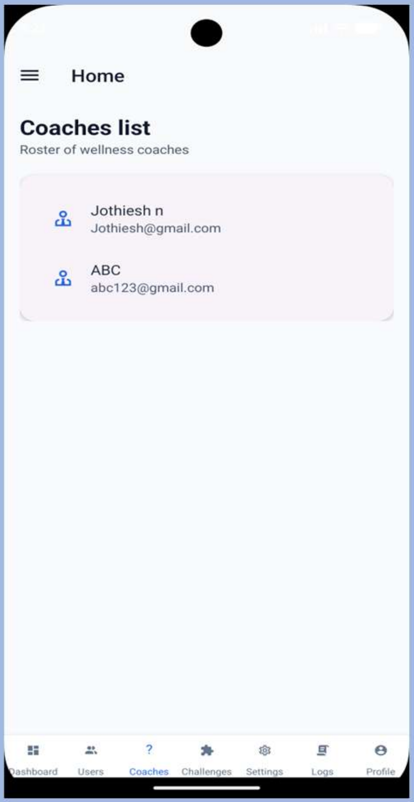


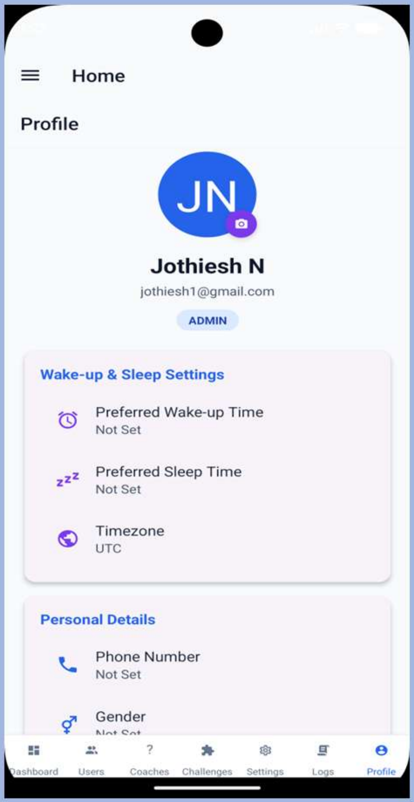


Admin Screens









Conclusion & Future Work

Conclusion

Successfully developed the Full Stack foundation of the Intelligent Cognitive Alarm Platform, including frontend, backend, authentication, user management, alarm scheduling, and dashboards.

Future Work

- Complete Habit Management
- Integrate AI-based Cognitive Challenges
- Perform end-to-end testing
- Deploy the application
- Optimize performance and user experience